UNIVERSITY OF PLYMOUTH

PUSL3190 Computing Project

Project Proposal

An Online Platform for Saltern Staff and Buyers

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Problem Statement

In Sri Lanka's economy, the salt industry supports the economy by supplying and exporting raw salt. The processes in local salt manufacturing remain mostly manual and on paper. Checking the salinity of the seawater, monitoring the crystallization process, planning maintenance schedules, tracking inventories, organizing transport, distribution, and sales are all activities recorded in books and spreadsheets. Since most of the data is recorded and filed away on paper, production, finance, and sales information is not integrated in any mapped-out unified system.

Inefficiencies, redundancy, and poor decision making stem from the paper trail. Real-time monitoring of stock levels, production costs, employee activities, and the myriad of other parameters is nearly impossible, let alone seamless.

In the absence of predictive analytics, unanticipated rain on seawater could affect production and profit margins. A saltern manager's reliance on manual weather tracking can diminish accuracy, but it is an accepted method of control in many factories.

From the customer perspective, the lack of a digital interface means there is limited engagement and no transparency. Buyers have to use the telephone or make a physical trip to the saltern to get information about the availability of a product, its price, or the expected delivery time. This is incredibly time consuming and inefficient. There is also no computerized order tracking, no digital invoicing, and no information about the delivery status of the order.

The absence of a centralized digital platform means several key problems will surface.

- There is no visibility of information which is vital for management decisions.
- Manual record keeping means there are errors and duplication.
- Reports on finances and products are late and poorly communicated among the several departments which are production, finance, transport, and sales.
- There is little or no communication with the customers about prices, product availability, and delivery status.
- Alerts needed for operational control are slow or inaccurate. Communication fails. These include the critical alerts of sudden changes in customer orders, adverse weather which could impact production, and increased operational failure.

These limitations can only be solved with an integrated online platform which will automate and digitalize the salt pan operation. This will allow collaboration with staff and customers in a seamless and improved operational flow.

Consequently, an integrated online interface to unify all saltern activities, including the collection of seawater and the selling of products, is necessary. This system will provide the ability to monitor information in real-time, communicate and collaborate with customers in an open manner, automate reporting, and create forecasting models. Such improvements will provide a considerable internal digital transformation, while also securing the longevity of the Sri Lankan salt industry regarding its competitiveness and sustainability.

Project Description

Overview

By combining all of the saltern's essential operations into a single, centralized digital system, this project seeks to design and develop an online platform for both slattern employees and clients. While the proposed platform will provide customers with an online portal to purchase salt, view product details, and track their orders, it will also enable staff to digitally record, monitor, and manage daily operations.

The platform will have two main user interfaces:

1. Staff Portal: to manage transportation, employee information, production, stock, and finances.
2. Customer Portal: for ordering, tracking deliveries, viewing available products, and paying online.

By replacing the current manual methods, this system will increase operational efficiency, accuracy, and transparency throughout the saltern's business workflow.

Project Objectives and Explanations

Main Objective

To create a web-based platform that effectively connects employees and clients by centralizing and automating all of a slattern's sales and operational procedures.

Specific Objectives

1. Digitize Operational Management - Automate quality control logs, stock updates, and salt production tracking.
2. Financial Management - Track profit and loss in real time, create financial reports, and keep track of expenses.
3. Human Resource Management - Control employee attendance, pay, and work distribution.
4. Customer Relationship Management - Let clients sign up, browse salt varieties, order, and pay online.
5. Logistics and Distribution - Monitor maintenance plans, delivery status, and transportation expenses.
6. Report Generation - Create analytical dashboards to support managerial choices.

Anticipated Results

- A decrease in human error and paperwork.
- Instant access to operational data.
- Better departmental communication.
- Increased trust and satisfaction among customers.
- Enhanced decision-making via automated analytics

Core features

1. Production yield and field management module for employees
2. Harvest logging - Offers field workers a web application interface to record harvest volumes and quality grades like edible, industrial, and separately in real time.

It offers automatically generated charts that show the yield obtained each month or year, giving users a sense of how much the yield has increased or decreased.

1. Ponds management module for staff

- Provides a web application interface to get an idea of the total ponds related to the salt marsh and the newly produced ponds.
- Develop a web interface to digitally log daily brine levels in ponds and gain insight into pond performance over time.
- Develop a web interface to log daily weather digitally.
- Weather prediction tool – Auto-generate charts to predict future weather using previous weather conditions

2. Harvest inventory management module for staff

- Real-time insight into production - Allow staff to know the exact quantity of each product batch in real time.
- Develop a web interface to digitally log the amount of product received in and the amount of product delivered out in real time.
- Inventory Alerts - Set up automatic alerts when inventory levels are below the required level or when capacity limits are reached.
- Auto-generate charts to gain a better understanding of inward vs. outward on a monthly and annual basis through inward and outward information, and increase industry profits.

3. Equipment Inventory Management Module for Staff

- Real-time insight into machines and equipment in the warehouse: Allow staff to know the exact quantity of each machine and equipment in real time.
- Set up the ability to report errors on existing machines and equipment.
- Set up the ability to update maintenance performed on existing machines and equipment and log information on machines and equipment that are removed from use.

4. **Expenses and Finance Module for staff**

- Create a web interface to easily process and view employee salaries.
- Create a web interface to easily process and view transportation expense reports.
- Create a web interface to easily process and view maintenance and repair cost reports.
- Create a dashboard to auto-generate monthly profit and loss reports in chart format for all relevant budgets.

5. **Admin dashboard**

- Approving orders
- Provides auto-generation charts to get an idea of the production Vs. sales.
- Providing access to relevant modules according to job role.

6. **Customer and Sales Portal for Customers**

- Self-service ordering - Enable customers to log in, view current, accurate product availability, and place new orders through online payments.
- Order Tracking - Enables customers to track the status of their order and gain insight into the status of their order.
- Order History - Makes it easy to access information about past invoices and purchases. Enables customers to provide reviews about their order and service.

Project Keywords

i. Saltern Management System
ii. Digital Transformation
iii. Enterprise Resource Planning
iv. Real-Time Analytics
v. E-commerce integration

Research Gap

Enterprise Resource Planning and Inventory Management Systems have been generalized and integrated into multiple business operations (Kumar & Saini, 2020; Tarigan et al., 2021). However, there is limited research and developed systems focused on salt operations (Rao et al., 2022; Silva & Perera, 2023). To this day, there are no modules that optimize and automate processes unique to salt pans, like tracking evaporation phases and managing data for pond-level brine concentration (Srinivasan & Ganesan, 2021). Systems that are limited in customization still do not integrate and automate decision-making processes for working during rainfall and predicting salt yield (Fernando & Jayasinghe, 2023).

Considering that the salt pans are comparatively lagging in digitization, the systems do not incorporate and automate the processes for the entire value chain, starting from seawater collection to distribution of the finished salt products (Wijesekera et al., 2024; Bhattacharya & Roy, 2021).

From a consumer viewpoint, there are no platforms that specifically allow customers to view the salt products and categories, check pricing and availability, place orders, pay, and track orders (Gunasekara & Ranasinghe, 2023). Systems do not integrate and automate invoice generation for completed orders (De Silva, 2022). This scenario limits transparency, and there are fewer opportunities for customers to engage with the business (Perera & Fernando, 2023).

In existing applications and research, the most notable absence, and digitization gap, is a complete beginning-to-end workflow (Jayawardena et al., 2024; Sharma et al., 2020).

Research gaps include:

- The absence of an online platform tailored to the salt pan industry (Silva & Perera, 2023).
- There is no integrated solution that combines customer interactions and staff operations (Rao et al., 2022).
- A poorly designed user interface (Jayawardena et al., 2024).
- Poor data visualization – the tools that are currently available lack visual analytics and dashboards to aid in managerial decision making (Tarigan et al., 2021).
- Limited predictive analytics – forecasting models for yield prediction, financial analysis, and maintenance planning are absent from current systems (Bhattacharya & Roy, 2021; Fernando & Jayasinghe, 2023).
- Insufficient integration with cloud-based solutions – remote access and real-time collaboration are not possible because most local salterns still keep their records offline (Wijesekera et al., 2024).
- Not using weather forecasting methods directly (Srinivasan & Ganesan, 2021).

This initiative aims to bridge these gaps and improve efficiency and transparency for all stakeholders by developing a comprehensive, customized web platform that encompasses the entire salt industry workflow (Perera & Fernando, 2023; Gunasekara & Ranasinghe, 2023).

Requirements Analysis

Technical Requirements

- Front-End:- React.js
- Back-End:- Node.js and Express.js
- Database:- Mongo DB
- Hosting:- Cloud-based deployment - AWS
- Payment Integration:- Online payment gateways like LankaQR, Pay Here
- Authentication and security :- Password hashing, JWT TOKEN, Role-based login as Admin, Staff, Customer
- Notification System:- Firebase Cloud Messaging , SendGrid
- Generate visual dashboards and reports:- libraries like Chart.js and Recharts
- Hardware:- Computers with internet access

Knowledge Requirements

- MERN stack as web development frameworks
- Database design and management
- API integration
- User Experience design principles
- Backup, Recovery and version control :- GitHub , Docker
- Research Knowledge:- Understanding of saltern operations, production stages, and business workflows. Awareness of how environmental factors like weather, brine concentration affect production

Finance (Budget Plan)

Items	Estimated Cost (LKR)
Annual web hosting and domain registration	8, 000
Field visits, documentation	5,000
printing, Communication	2,000
Total Estimated Cost	LKR 15 ,000

External Organizations and Supporting Staff

External Parties

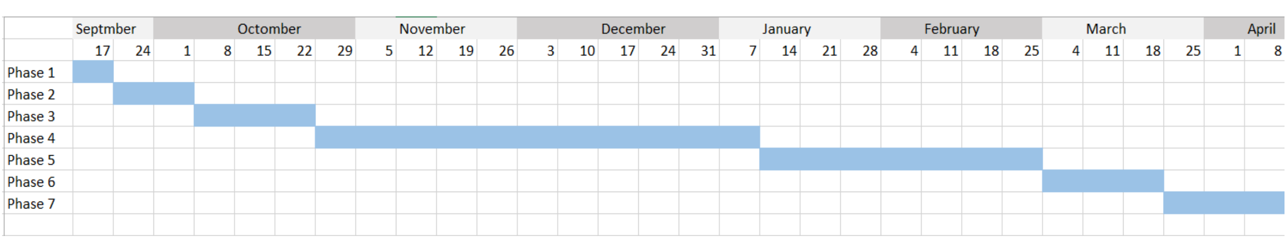
- Local Saltern Management - To provide real-world data and process insights.
- ICT Supervisors - For technical guidance and review.

Supporting Staff

- Field Supervisors - For providing process data.
- Financial Officer - For verifying accounting records.

Time Frame

Task	Task Description	Start Date	End Date	Duration	Dependencies	Milestones
Phase 1	Proposal approval and Project Initiation	17/09/2025	24/09/2025	1 week	None	Project approved
Phase 2	Detailed requirement documentation, complete process flow mapping, System Architecture Plan.	24/09/2025	08/10/2025	2 weeks	Proposal approval	Plan finalized
Phase 3	UI/UX design, interface prototypes, database schema finalized	08/10/2025	29/10/2025	3 weeks	Detailed requirement documentation, complete process flow mapping, System Architecture Plan.	UI/UX design finalized
Phase 4	Backend development	29/10/2025	14/01/2025	11 weeks	UI/UX design	
Phase 5	Frontend development	14/01/2025	04/03/2026	7 weeks	UI/UX design	
Phase 6	Internal QA, Bug Fixing	04/03/2026	25/03/2026	3 weeks	Development	Finalize system
Phase 7	final documentation, and report submission.	25/03/2026	08/04/2026	2 weeks	Internal QA and Bug Fixing	Final report submit
Total Duration				29 weeks (7months)		



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